R Code

```r

################################################################
# PRACTICAL 7: ARIMA MODELS IN R
# Dataset: nottem
################################################################

# Objective:
# To study ARIMA models, perform differencing,
# fit an ARIMA model, and examine residuals.

# ============================================================
# 1. LOAD THE DATASET
# ============================================================

data(nottem)

# Display observations
nottem

# ============================================================
# 2. BASIC INFORMATION
# ============================================================

class(nottem)
length(nottem)
start(nottem)
end(nottem)
frequency(nottem)

# ============================================================
# 3. PLOT THE ORIGINAL TIME SERIES
# ============================================================

plot(nottem,
     main = "Monthly Mean Air Temperature in Nottingham",
     xlab = "Year",
     ylab = "Temperature (Fahrenheit)")

# ============================================================
# 4. ACF AND PACF OF ORIGINAL SERIES
# ============================================================

acf(nottem, main = "ACF of nottem")

pacf(nottem, main = "PACF of nottem")

# ============================================================
# 5. FIRST DIFFERENCE
# ============================================================

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d_nottem <- diff(nottem)

plot(d_nottem,
     main = "First Difference of nottem",
     xlab = "Year",
     ylab = "Change in Temperature")

acf(d_nottem,
    main = "ACF of First-Differenced Series")

pacf(d_nottem,
     main = "PACF of First-Differenced Series")

# ============================================================
# 6. SEASONAL DIFFERENCE
# ============================================================

# Monthly observations have a seasonal period of 12.

seasonal_nottem <- diff(nottem, lag = 12)

plot(seasonal_nottem,
     main = "Seasonal Difference of nottem",
     xlab = "Year",
     ylab = "Seasonal Difference")

acf(seasonal_nottem,
    main = "ACF of Seasonal-Differenced Series")

# ============================================================
# 7. FIT ARIMA(1,1,1)
# ============================================================

# ARIMA(p,d,q):
# p = autoregressive order
# d = number of regular differences
# q = moving-average order

model1 <- arima(nottem,
                order = c(1, 1, 1))

# Display model output
model1

# Estimated coefficients
model1$coef

# AIC value
AIC(model1)

# ============================================================
# 8. FIT ARIMA(1,0,1)
# ============================================================

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# This model is appropriate only if the original
# series is adequately stationary.

model2 <- arima(nottem,
                order = c(1, 0, 1))

model2

# Compare AIC values
AIC(model1)
AIC(model2)

# ============================================================
# 9. AUTOMATIC SEASONAL ARIMA MODEL
# ============================================================

# Install only once if required:
# install.packages("forecast")

library(forecast)

seasonal_model <- auto.arima(
  nottem,
  seasonal = TRUE
)

# Display selected model
seasonal_model

# Display ARIMA orders
arimaorder(seasonal_model)

# ============================================================
# 10. RESIDUAL DIAGNOSTICS
# ============================================================

checkresiduals(seasonal_model)

# ============================================================
# 11. FORECASTING
# ============================================================

# Forecast the next 12 months

forecast_values <- forecast(
  seasonal_model,
  h = 12
)

# Display forecast
forecast_values

# Plot forecast
plot(forecast_values,
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      main = "Forecast of Monthly Nottingham Temperatures",
      xlab = "Year",
      ylab = "Temperature (Fahrenheit)")


```